Proof-of-Concept Lesional Skin Classification with IDC Wearable Sensors under Data Scarcity

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Continuous monitoring of atopic dermatitis (AD) with wearable interdigitated capacitive (IDC) sensors offers a non-invasive proxy for skin barrier integrity, but public data remain scarce. The available corpus contains 13 patients, each with one 30-second lesional trace and one 30-second non-lesional trace. This report evaluates whether short trace prefixes contain enough discriminative signal and whether models can generalize to unseen patients without relying on a population-wide threshold. Six raw-trace statistical features normalized within each leave-one-subject-out (LOSO) fold yield a LinearSVC with GroupKFold AUC 0.822 and correct pairwise ranking for all 13 held-out patients. A compact 1D CNN with Global Average Pooling slightly improves GroupKFold AUC to 0.871, suggesting that both models mainly use per-site mean capacitance rather than complex temporal structure. Self-supervised pretraining on WESAD electrodermal activity (EDA) data followed by frozen-layer transfer reaches GroupKFold AUC 0.827, which is above the SVM but below the from-scratch CNN, indicating that the EDA-to-IDC sensor-physics gap limits transfer at N=26 traces. The window-length experiment shows that the apparent advantage of early prefixes is driven by missing post-10 s samples in S-02, S-03, and S-04 that become constant after forward-fill preprocessing, not by evidence that shorter measurements are clinically superior. Overall, the results support personalized intra-individual AD tracking, while population-level validation requires a larger and cleaner IDC dataset.

Additional Key Words and Phrases: atopic dermatitis, interdigitated capacitive sensor, data scarcity, time-series classification

1 Motivation

Atopic dermatitis (AD) is a common chronic inflammatory skin disease, affecting approximately 15–20% of children and 2–10% of adults worldwide [5]. It is marked by impaired skin-barrier function, itch, and recurrent flares, so disease severity can change substantially between clinic visits. Clinical assessment also depends on observer judgment. The Eczema Area and Severity Index (EASI), for example, has only moderate inter-observer reliability [6]. These limitations motivate continuous, at-home monitoring.

Wearable systems for AD commonly measure behavior, such as scratching, or skin barrier function. Behavioral sensing is useful but indirect. In contrast, capacitance-based barrier sensing targets the physiological state of the skin. This report focuses on the interdigitated capacitive (IDC) sensor introduced by Todorov et al. [8]. The device measures dielectric permittivity at the skin interface, providing a proxy for hydration and barrier integrity. It samples capacitance at 1 Hz over a 30-second contact window. Todorov et al. showed that non-lesional skin usually has higher capacitance than lesional skin within the same patient.

The central limitation is data availability. To my knowledge, the Todorov et al. corpus is the only public IDC dataset with AD lesional labels. It includes 13 patients, each with one lesional and one non-lesional trace, for 26 total traces. This small sample makes ordinary supervised learning fragile and makes patient-level leakage a major risk.

Two questions guide the experiments. First, is a short trace prefix sufficient for classification, or does the full 30-second window improve discrimination? Todorov et al. reported early contact-settling drift but did not test shorter windows. Second, can a model trained on other patients correctly rank the lesional and non-lesional sites for a held-out patient without using a population-wide threshold? This directly tests the intra-individual tracking use case emphasized by Todorov et al.

2 Related Work

AD wearable sensing

Khan et al. [6] review AD wearables across behavioral/itch sensing, surface barrier-function sensing, and intradermal sensing. The IDC sensor studied here belongs to the surface barrier-function category, which is more directly tied to skin pathology than scratch behavior. Because few systems release labeled public data, the Todorov corpus is unusually valuable despite its small size.

IDC sensors and skin capacitance

Todorov et al. [8] introduced a printed IDC sensor for wearable AD monitoring and showed that non-lesional capacitance exceeds lesional capacitance at the individual-patient level. They also reported Corneometer correlations, supporting capacitance as a hydration proxy. Their results suggest that a population-wide threshold is unreliable because baseline capacitance varies across patients. This project formalizes that issue with leave-one-subject-out validation and GroupKFold AUC.

Machine learning on small AD datasets

Xing et al. [9] provide a useful small-data comparison from nocturnal scratch detection. In a seven-subject wrist-actigraphy dataset, a feature-engineered gradient-boosting model outperformed a 1D CNN on minority-class F1 (0.45 vs. 0.39). Although the sensing modality differs from IDC capacitance, the result supports the same methodological caution: with very small AD datasets, lower-capacity feature-based models can match or exceed end-to-end deep learning.

3 Project Aim

This work has three aims within the data constraint of the Todorov et al. [8] corpus. All results are proof-of-concept findings on N=13 patients, not clinical validation.

(1) **Replication.** Reproduce the per-patient non-lesional > lesional capacitance ordering from Todorov et al. Fig. 9(a).
(2) **Cross-patient generalization.** Quantify the difference between intra-individual pairwise ranking and population-level threshold accuracy, using GroupKFold AUC as the main discriminability metric.

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(3) **Learning under scarcity.** Compare a six-feature LinearSVC with a compact 1D CNN and test whether self-supervised EDA pretraining improves IDC classification at N=26 traces.

4 Methodology

4.1 Dataset

The dataset comes from Todorov et al. [8]. Thirteen AD patients each contributed two 30-second IDC capacitance traces sampled at 1 Hz: one lesional site and one non-lesional site, for 26 traces total. A raw-data audit shows that exactly three patients, S-02, S-03, and S-04, have incomplete IDC recordings. For these patients, both lesional and non-lesional traces contain valid samples only from 1 to 10 seconds, followed by missing entries from 11 to 30 seconds. Thus, the missingness affects six of the 26 IDC traces.

The preprocessing pipeline forward-fills missing sensor entries. Consequently, for S-02, S-03, and S-04, the final 20 seconds become repeated copies of the 10-second value rather than genuine measurements. This data-completeness issue is central to the window-length analysis (Section 5.4).

4.2 Evaluation Framework

All experiments use patient-safe evaluation. *LOSO* has 13 folds, with one patient held out per fold. Since each test fold has one lesional and one non-lesional sample, LOSO AUC reduces to a binary pairwise-ranking indicator and should not be compared with ordinary multi-sample AUC values. *GroupKFold* (n=5) places 2–3 patients in each test fold, producing a graded fold-level AUC with lower variance. This is the main discriminability metric. *Pooled-LOSO AUC* concatenates all LOSO test scores into one ROC curve and is reported only as a reference because pooled leave-one-out AUC is negatively biased [1]. All scalers and baseline estimators are fit only on training-fold data.

When patient-baseline centering is evaluated under strict cross-patient testing, the held-out patient receives a training global-mean fallback. This choice is intentionally conservative because it prevents the held-out patient's baseline from entering the fitted normalization parameters. It is not proposed as the best deployment strategy. In a real paired-site measurement setting, a new patient's own unlabeled lesional and non-lesional candidate traces could be centered together before classification.

4.3 Feature Extraction (Stage 2)

Each trace is represented by six summary features: mean, standard deviation, median, range (max−min), slope from ordinary least squares regression over time, and skewness. Features are computed from raw traces because trace-level z-score normalization would force all trace means to zero and remove the highest-importance features. Random Forest feature importances are reported in Table 2 and Figure 2.

4.4 Normalization Comparison (Stage 3)

Cross-validation estimates can be biased when normalization is fit on the full dataset [4]. Therefore, all feature scalers and patient-baseline estimators are fit within the training fold only. Five strategies are compared: feature-level z-score, feature-level RobustScaler, fold-safe patient-baseline centering, trace-level z-score, and trace-level min-max. GroupKFold AUC is used for the main comparison (Table 3).

4.5 Window Length Experiment

Sixteen window configurations are evaluated: prefix windows (0→5, 10, 15, 20, 25, 30 s) and post-settling windows (5→N and 10→N s). For each window, features are re-extracted and a LinearSVC with feature-level z-score normalization is evaluated under LOSO. The goal is to test whether early contact-settling drift dominates classification or whether later samples improve discrimination. The forward-filled traces confound this test, as discussed in Section 5.4.

4.6 From-Scratch 1D CNN (Track A)

Track A tests whether end-to-end temporal modeling adds value beyond the six handcrafted features. The CNN input is the raw 30-sample IDC trace after fold-safe standardization by the training-set global mean and standard deviation. TinyCNN uses two one-dimensional convolutional layers with kernel size 3 and padding 1, followed by Global Average Pooling, Dropout (0.3), and a two-class linear output layer. There is no max-pooling layer, so the temporal length remains 30 until the final average-pooling step. This design keeps the model small enough for the N=26 trace setting while still allowing local temporal filters before the mean-like pooling operation.

Models are trained with Adam (lr = 0.001) for 300 epochs. A jitter variant adds training-only Gaussian noise with $\sigma = 0.2$ on the standardized input scale. Both variants are evaluated with five random seeds. For each seed and each fold, a fresh model is trained from scratch so that test traces from the held-out patient are never used during fitting.

Table 1. TinyCNN architecture used in Track A. Shapes omit the batch dimension. The model has 466 trainable parameters.

Layer	Output shape	Parameters	Role
Input trace	1 × 30	0	Fold-standardized IDC sequence
Conv1d + ReLU, 1 → 8, k=3	8 × 30	32	Local temporal filtering
Conv1d + ReLU, 8 → 16, k=3	16 × 30	400	Higher-level 1D feature maps
Global average pool	16	0	Averages each feature map over time
Dropout, p=0.3	16	0	Regularizes the classifier head
Linear classifier, 16 → 2	2	34	Lesional vs. non-lesional logits

4.7 Self-Supervised Transfer Learning (Track B)

Track B tests whether a larger external skin-signal dataset can improve learning from 26 IDC traces. Following the transfer-learning motivation of Fawaz et al. [2, 3], the TinyCNN encoder is paired with a symmetric 1D convolutional decoder to form an autoencoder. The autoencoder is pretrained on 2,889 non-overlapping 30-sample WESAD EDA windows [7]. WESAD wrist EDA is first downsampled from 4 Hz to 1 Hz by block averaging, segmented into 30-second windows, and filtered to remove constant windows.

Before MSE reconstruction pretraining, the WESAD windows are globally z-scored using the mean and standard deviation computed across all WESAD windows. This normalization prevents the autoencoder from reconstructing raw EDA amplitudes in microsiemens, but it is not per-window normalization and it does not use IDC capacitance statistics. Therefore, Track B is transfer from globally normalized EDA dynamics to fold-standardized IDC traces, not transfer from raw EDA amplitude to raw IDC capacitance. No WESAD stress or affect labels are used. After pretraining, the decoder is discarded, the encoder is frozen, and only the classification head is trained on IDC traces under the same LOSO and GroupKFold protocols as Track A. The autoencoder mirrors the Track A encoder with two decoder convolutions, $16 \longrightarrow 8$ and $8 \longrightarrow 1$, both using kernel size 3 and padding 1. During transfer, the 432 encoder parameters are frozen and only the 34-parameter classification head is optimized. The full pretraining and evaluation pipeline is repeated across five seeds.

5 Results

5.1 Stage 1: Replication of Todorov Fig. 9(a)

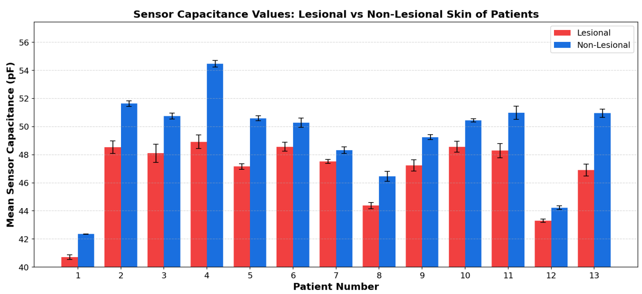

Fig. 1. Per-patient mean capacitance for lesional and non-lesional sites (13 patients). Non-lesional exceeds lesional for every patient, replicating Todorov et al. [8] Fig. 9(a). Absolute levels vary by ~10 pF across patients, explaining why a population-wide threshold cannot be defined.

Non-lesional capacitance exceeds lesional capacitance for every patient, matching Todorov et al. Fig. 9(a). The within-patient gap is consistent (2–5 pF), but absolute baselines vary by ≈10 pF across individuals. This explains why a single population-wide threshold is unreliable.

5.2 Stage 2: Feature Importance

Mean and median are the two strongest Random Forest features, which is consistent with the per-patient capacitance gap in Stage 1. Slope ranks third, suggesting that linear drift over the 30-second contact window adds some information beyond level alone. All six raw-trace features remain non-zero in importance.

5.3 Stage 3: Normalization Comparison

Feature-level z-score and RobustScaler tie on every metric, so IQR-based scaling adds no practical benefit at N=26. Trace-level z-score performs near chance because it removes the mean and median signals. Patient-baseline centering gives perfect LOSO pairwise

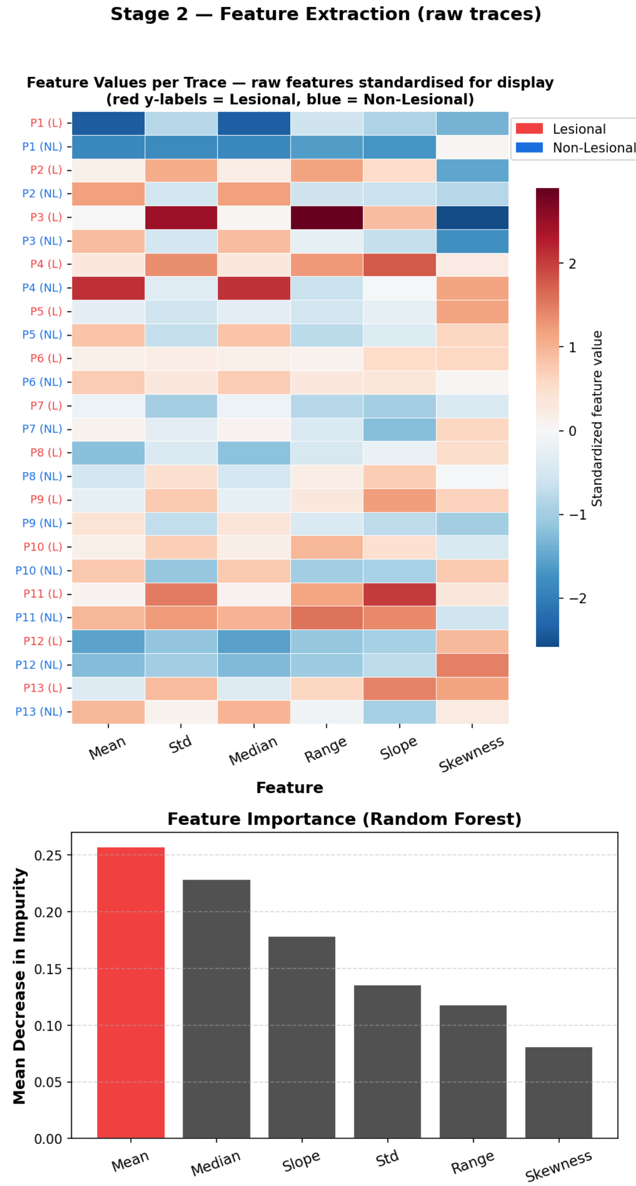

Fig. 2. Random Forest feature importances (top) and per-feature LOSO box plots (bottom) for all six statistical features extracted from raw 30-second traces (N=26).

ranking, but that value is a two-sample within-patient ranking result rather than an ordinary AUC estimate. Under GroupKFold, its AUC is 0.817, slightly below feature z-score (0.822). Feature-level z-score is therefore used in later stages.

5.4 Stage 4: Window Length

The 0–5 s prefix has the highest AUC (0.923), while performance decreases as the window extends. Post-settling windows perform much worse, with post-10 s classification near random. This is a

Table 2. Random Forest feature importances on raw traces (N=26, `random_state=42`).

Rank	Feature	Importance
1	Mean	0.257
2	Median	0.229
3	Slope	0.179
4	Std	0.136
5	Range	0.118
6	Skewness	0.081

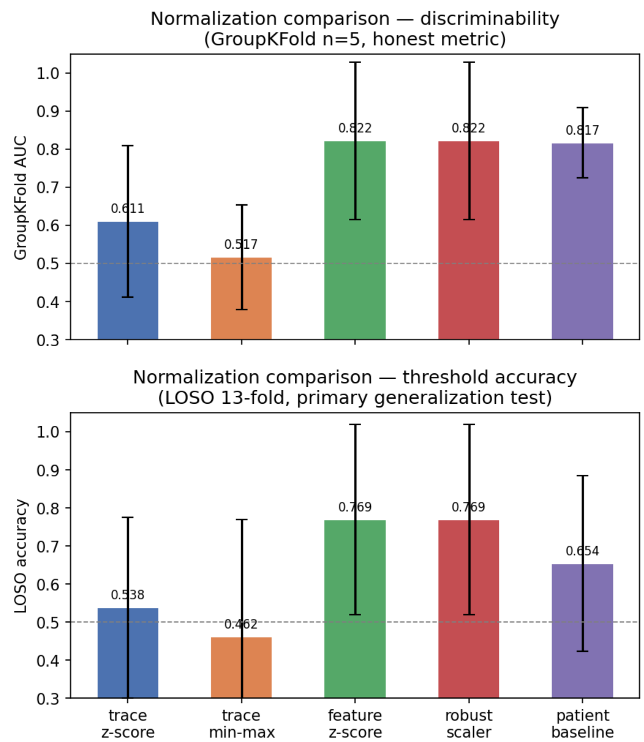

Fig. 3. GroupKFold AUC and LOSO accuracy for five normalization methods (LinearSVC). Feature z-score and RobustScaler tie on every metric. Patient-baseline centering gives perfect LOSO pairwise ranking, but falls to 0.817 under GroupKFold.

data-quality confound, not a clinical finding. The raw audit shows that S-02, S-03, and S-04 contain valid IDC samples only through 10 seconds in both lesional and non-lesional traces. After forward-fill preprocessing, later samples in these six traces are mostly constant. The experiment therefore shows that the current dataset is most complete in the first 10 seconds, not that shorter measurements are clinically better.

5.5 Stage 5: Evaluation Expansion

The model correctly ranks the lesional site for all 13 held-out patients, directly supporting intra-individual tracking. LOSO threshold accuracy is lower (0.654), which reflects the lack of a reliable population-wide decision boundary. GroupKFold AUC (0.817) and pooled-LOSO AUC (0.775) provide additional discriminability checks. The accuracy–ranking gap also argues against simple data leakage.

5.6 Stage 6A: CNN Track A vs. SVM

The CNN-scratch train–test accuracy gap is 0.077, below the pre-registered 0.20 overfitting trigger. Its GroupKFold AUC is 0.871, above the SVM reference (0.822). The jitter variant gives the highest GroupKFold AUC, increasing from 0.871 to 0.889. This ≈0.018 gain is plausible but modest. Because the dominant signal is a level difference, Gaussian jitter likely discourages the CNN from memorizing exact fold-specific capacitance levels and encourages a smoother decision boundary around the mean-capacitance signal. It should therefore be interpreted as regularization of the same level-based representation, not as evidence that noise augmentation reveals a clinically meaningful waveform pattern.

5.7 Stage 6B: Transfer Learning (Track B)

The transfer CNN reaches GroupKFold AUC 0.827, slightly above the SVM but below both from-scratch CNN variants. Its train-test gap is only 0.011, so the issue is not overfitting. Rather, the frozen EDA-pretrained encoder does not provide a better IDC representation. Across all models, the central signal remains per-site mean capacitance.

6 Discussion

6.1 Why the CNN Ties the SVM

The main Stage 6 result is the small gap between the CNN and the feature-based SVM. Global Average Pooling averages the learned 1D feature maps over time, so the CNN can easily reproduce a mean-level representation. Because a model with the full raw trace only slightly outperforms a model with six summary features, the useful signal in this corpus appears to be a level difference rather than a temporal pattern. Architectures with implicit or explicit mean pooling are likely to reach a similar ceiling on this dataset.

6.2 Data Quality Confound

The window-length experiment is limited by a specific data-completeness artifact in the IDC sensor files. The raw CSV audit shows that exactly three patients, S-02, S-03, and S-04, have valid IDC sensor values only through 10 seconds in both lesional and non-lesional recordings. The entries from 11 to 30 seconds are missing in the raw files. After forward-fill preprocessing, the post-10 s region for these six traces becomes artificially constant.

For this reason, the post-10 s window result should not be interpreted as evidence that late IDC measurements are physiologically uninformative. Instead, it shows that late-window classification is unreliable when part of the dataset contains missing values that are converted into flat repeated segments. The stronger conclusion is that the usable signal in the current dataset is concentrated in the first 10 seconds because those samples are complete across all 26 IDC traces. A future analysis should repeat the window-length experiment after either removing incomplete traces or recollecting full 30-second IDC recordings for S-02, S-03, and S-04.

Table 3. Normalization comparison for LinearSVC with feature-level normalization, LOSO and GroupKFold (n=5). GroupKFold AUC is the main discriminability metric. † marks a pairwise LOSO ranking value, not an ordinary multi-sample AUC advantage.

Method	LOSO Acc	LOSO F1	LOSO Rank†	Pooled-LOSO AUC	GKF Acc	**GKF AUC**
Feature z-score (primary)	0.769 ± 0.249	0.744 ± 0.350	0.846	0.781	0.750 ± 0.167	**0.822**
RobustScaler	0.769 ± 0.249	0.744 ± 0.350	0.846	0.793	0.750 ± 0.167	**0.822**
Patient baseline (fold-safe)	0.654 ± 0.231	0.462 ± 0.444	1.000†	0.775	0.700 ± 0.113	0.817
Trace z-score	0.538 ± 0.237	0.410 ± 0.396	0.769	0.615	0.583 ± 0.105	0.611
Trace min-max	0.462 ± 0.308	0.308 ± 0.402	0.462	0.379	0.433 ± 0.178	0.517

Table 4. Selected window-length results (LinearSVC, feature z-score per segment, LOSO). Full table: `results/stage4_window_length.csv`.

Family	Window	Acc	AUC
Prefix (0⟶N)	0–5 s	0.731 ± 0.317	0.923 ± 0.267
	0–10 s	0.731 ± 0.317	0.846 ± 0.361
	0–30 s	0.538 ± 0.237	0.769 ± 0.421
Post-5 s (5⟶N)	5–30 s	0.231 ± 0.317	0.154 ± 0.361
Post-10 s (10⟶N)	10–30 s	0.346 ± 0.303	0.192 ± 0.369

Table 5. Evaluation metrics for LinearSVC with fold-safe patient-baseline normalization, full 30 s traces.

Method	Acc	F1	GKF AUC
LOSO (13 folds, primary)	0.654 ± 0.231	0.462 ± 0.444	N/A
GroupKFold (n=5)	0.700 ± 0.113	0.651 ± 0.146	0.817 ± 0.092

Pooled-LOSO AUC = 0.775 (negative bias per [1]).

6.3 Why Transfer Learning Did Not Help

The weak transfer-learning result is unlikely to be explained only by raw amplitude mismatch. In Track B, WESAD EDA windows are globally z-scored before MSE autoencoder pretraining, and IDC traces are standardized within each evaluation fold before downstream classification. However, these normalization steps do not remove the deeper modality mismatch. EDA measures skin conductance linked partly to eccrine sweat activity, while IDC measures capacitance or permittivity-related response at the stratum corneum barrier. Their physical meaning, amplitudes, temporal dynamics, and likely artifacts differ.

Track B also freezes the pretrained convolutional encoder. If the autoencoder learns EDA-specific drift or artifact structure, the frozen encoder cannot adapt fully to the IDC target task. Since the IDC classification signal is dominated by a simple level difference, pretrained EDA waveform features provide little advantage over a CNN trained directly on the IDC traces. Future work could test unfrozen fine-tuning or pretraining on a larger IDC-specific source dataset.

6.4 Clinical Interpretation

The strongest result of this study is not population-level diagnosis but within-patient ranking. In every patient, the non-lesional site has higher mean capacitance than the lesional site, and the best fold-safe models largely preserve this ordering for held-out patients. This suggests that IDC sensing is better suited for personalized monitoring than for a universal diagnostic threshold. A practical wearable workflow would compare a patient's current skin sites against the same patient's paired or prior measurements rather than classify each trace against a fixed population cutoff.

This distinction matters because AD severity and skin capacitance are heterogeneous across patients. Skin thickness, hydration, sensor contact, and baseline barrier properties can shift absolute capacitance. These factors make a single cutoff fragile at small sample size. Repeated within-patient IDC measurements could still be useful for tracking flare progression, treatment response, or recovery of the skin barrier, but that use case requires complete recordings and larger longitudinal validation.

6.5 Limitations

Several limitations bound these findings. The dataset has only 13 patients, so LOSO fold-to-fold variance is high and all metrics should be treated as indicative. Six of the 26 IDC traces contain missing post-10 s samples, which limits conclusions about the optimal contact-window duration. The Corneometer correlation analysis was not performed, leaving the relationship between IDC features and established hydration measurements uncharacterized. TEWL measurements are excluded because of limited data availability. Tiny-CNN is one architecture without a hyperparameter search. Track B uses frozen convolutional layers only, so a two-phase pretrain-then-fine-tune strategy may produce different results. Finally, LOSO binary-fold AUC is not comparable to AUC values from ordinary multi-sample cross-validation.

7 Conclusion

This report presents a proof-of-concept pipeline for classifying lesional and non-lesional AD skin from a severely limited IDC wearable dataset. The replication stage confirms that non-lesional capacitance exceeds lesional capacitance within each patient, but patient baselines vary enough to make a single population-wide threshold unreliable. A LinearSVC using six raw-trace statistical features ranks the lesional site correctly for all 13 held-out patients and achieves GroupKFold AUC 0.822. A compact 1D CNN improves GroupKFold AUC to 0.871, but its small margin over the SVM suggests that the useful signal is mainly mean capacitance rather than temporal waveform structure. Transfer learning from WESAD EDA reaches GroupKFold AUC 0.827, showing limited benefit because EDA and IDC measure different skin properties. The window-length experiment further shows that apparent short-window advantages are driven by missing post-10 s samples that become constant after

Table 6. Full model comparison for Stages 6A and 6B. Values are mean±std over 5 random seeds. Track B GKF AUC (0.827) falls below both Track A variants despite domain-adjacent pretraining.

Model	Train Acc	LOSO Acc	GKF AUC	Pooled-LOSO
CNN-jitter	0.844 ± 0.002	0.777 ± 0.015	**0.889** ± 0.014	0.815 ± 0.021
CNN-scratch	0.846 ± 0.001	0.769 ± 0.000	0.871 ± 0.022	0.808 ± 0.024
CNN-transfer	0.726 ± 0.036	0.715 ± 0.039	0.827 ± 0.022	0.773 ± 0.008
SVM (ref.)	N/A	0.769	0.822	0.781

Track B LOSO rank = 0.985 ± 0.031, Train-test gap = 0.011.

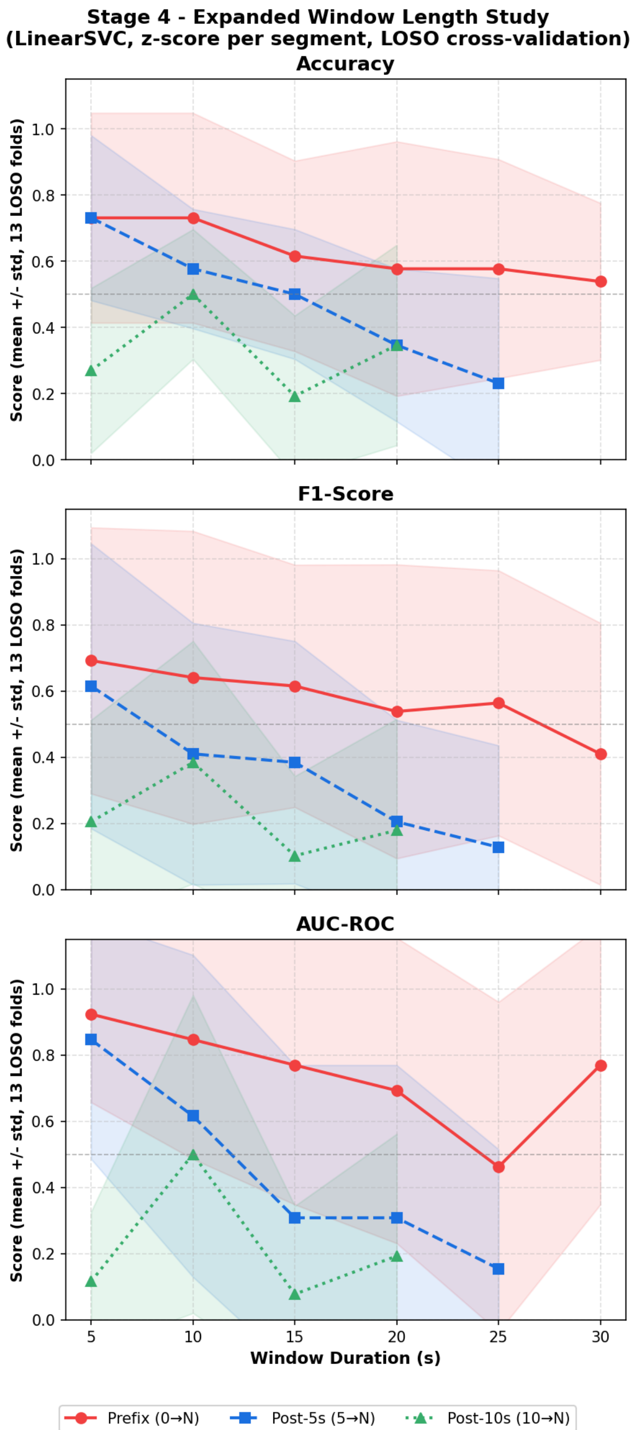

Fig. 4. LOSO AUC (mean ± std) for prefix and post-settling window families. Performance degrades sharply in post-10 s windows because S-02, S-03, and S-04 have missing raw samples after 10 s that become constant after forward-fill preprocessing.

forward-fill preprocessing. Overall, IDC sensing appears promising for personalized AD tracking, but robust clinical deployment will require larger, complete, artifact-controlled datasets.

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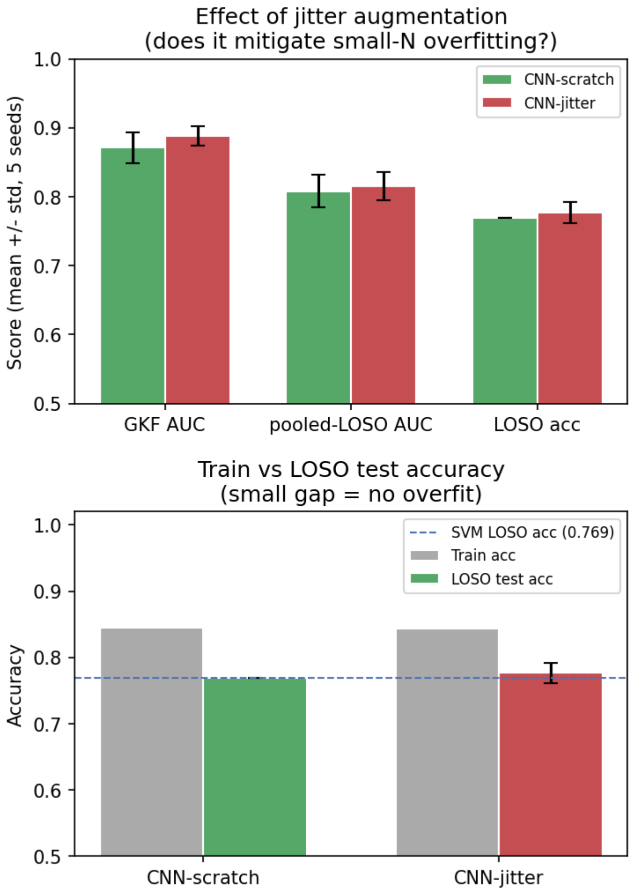

Fig. 5. Stage 6 Track A. *Top:* training vs. test accuracy per LOSO fold (5 seeds, CNN-scratch and CNN-jitter). Small train–test gap confirms no overfitting. *Bottom:* GKF AUC comparison (CNN-scratch, CNN-jitter, SVM reference).

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